

TAZNIA TABASSUM ONON

Software Engineer | Machine Learning | UI/UX & Graphic Designer

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🐙 GitHub 🌐 Portfolio 🔗 LinkedIn

OBJECTIVE

A dedicated Computer Science and Engineering student with a strong foundation in software development, web and backend technologies, application development, and machine learning. Experienced in developing software solutions, working with modern development tools and technologies, and building intelligent applications. Passionate about solving real-world problems through technology, with strong problem-solving, technical, and collaborative skills.

EDUCATION

BRAC University

Bachelor of Science in Computer Science and Engineering
– **CGPA:** 3.78 / 4.00

Dhaka, Bangladesh

2022 – 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Operating Systems, Embedded Systems, Computer Vision, Software Engineering
- **Undergraduate Thesis:** ADECS: An Agentic AI Framework for Anomaly Detection and Intelligent Decision Support in Elderly Care System (Grade: A+, Marks: 98.875/100)

Motijheel Ideal School and College

Higher Secondary Certificate (Science)
– **GPA:** 5.00 / 5.00

Dhaka, Bangladesh

2019 – 2021

Ideal School and College (Mugda Branch)

Secondary School Certificate (Science)
– **GPA:** 4.94 / 5.00

Dhaka, Bangladesh

2018 – 2019

ENTREPRENEURIAL EXPERIENCE

Co-Founder & Chief Operating Officer (COO)

Onnesok | www.onnesok.com

Jan 2025 – Present

Bangladesh

- Co-founded an innovation-driven tech startup focused on digital products, web tools, and game development.
- Direct strategic planning, tech roadmap execution, and product operations to scale creative concepts.

PART-TIME & PROFESSIONAL EXPERIENCE

Graphic Designer

Intellectium (Contract)

Jan 2024 – Dec 2025

Bangladesh

- Designed creative visual content for digital platforms, branding, and marketing campaigns.
- Collaborated with teams to deliver high-quality, engaging design solutions aligned with client needs.

Facilitator

Ecollab International

Sep 2024 – Jun 2025

International

- Conducted training sessions and workshops on leadership, communication, and collaboration.
- Mentored participants and supported program activities to ensure effective learning outcomes.

Coordinator

BuReD (BRAC University Research and Development Club)

Jan 2025 – Feb 2026

Dhaka, Bangladesh

- Coordinated club activities, workshops, and seminars aimed at promoting academic research.
- Facilitated student research initiatives, collaborating with teams and faculty mentors to support project execution.

Team Lead – Social Media & Content

Project Proyash (contract)

May 2024 – Jun 2025

Dhaka, Bangladesh

- Led the social media and content team, developing strategies to enhance online presence.

PROJECTS

Mobile & Web Applications | *Django, Flutter, Firebase, REST APIs*

- Engineered full-stack applications (**Learners, HireEasy**) with Flutter UI and Django backend.
- Implemented secure RESTful APIs, database schemas, user authentication, and Firebase cloud services.

Game Development | *Godot, Flutter, UI/UX*

- Developed interactive 2D/3D games (**ShooterX, Sky Ball, Adventure of Sui**) using Godot and Flutter.
- Engineered responsive game physics, dynamic animations, and intuitive player UI/UX interfaces.

AI-Powered Bionic Arm with Sensory Feedback | *Python, ML, Embedded Systems*

- Built an AI-driven prosthetic arm with real-time sensory feedback and embedded microcontroller logic.
- Integrated lightweight machine learning models for adaptive and accurate grasp pattern recognition.

WildTrackID: Deep Learning Wildlife Identification | *PyTorch, Swin Transformer, ArcFace*

- Created a metric-learning framework combining MegaDescriptor and Swin Transformer backbones.
- Applied ArcFace and cosine-similarity retrieval to accurately identify individual wildlife from image data.

Quadruped Robot (AI Robotic Dog) | *Python, Reinforcement Learning, MuJoCo*

- Simulated a four-legged robotic dog in MuJoCo physics engine for stable dynamic locomotion.
- Trained reinforcement learning policies to optimize terrain balance and autonomous obstacle navigation.

Reinforcement Learning Simulations | *Python, Q-Learning, OpenAI Gym*

- Built and benchmarked reinforcement learning agents across custom and standard OpenAI Gym environments.
- Implemented Q-Learning and Deep Q-Networks (DQN) to solve sequential decision-making challenges.

Face Detection System | *Python, OpenCV*

- Developed a real-time computer vision system for low-latency facial detection and multi-target tracking.
- Optimized frame processing pipelines for high accuracy under complex and low-light environments.

TECHNICAL SKILLS

Programming Languages: Python, Dart, C, C++, JavaScript, HTML/CSS, SQL

Frameworks & Backend: Django, Django REST Framework, Flutter, Android App Development

AI, ML & Computer Vision: Machine Learning, OpenCV, OpenAI Gym, Reinforcement Learning, MuJoCo

Game Development & Design: Godot, Blender, UI/UX Design, Graphic Design

Robotics & Systems: ROS, Robotics, Arduino, Embedded Systems

Tools & Databases: Firebase, SQLite, PostgreSQL, Git, GitHub

HONORS ACHIEVEMENTS

- **Gold Award** — IEEE R10 Humanitarian Technology Challenge (HTC) for Bionic Arm with Sensory Feedback, Oct 2024.
- **VC Special Recognition Award** — BRAC University, Fall 2025.
- **Seed Grant Recipient (90,000 BDT)** — University Innovation Hub Program, Sep 2026.
- **Merit-Based Scholarship** — BRAC University, awarded for academic excellence.
- **2nd Runner-Up** — IoTrix: National-Level IoT Embedded Systems Hackathon, Nov 2025.
- **2nd Runner-Up** — bdapps Innovation Summit (Project: ShooterX), Aug 2025.
- **1st Runner-Up** — Poster Presentation (GenesisX) at BRAC University Research for Development (BuReD), Dec 2024.
- **Pre-Seed Grant Recipient (55,000 BDT)** — University Innovation Hub Program, Mar 2025.
- **Honourable Mention** — International Science and Invention Fair (ISIF Bangladesh), Oct 2025.
- **Best Campus Ambassador** — Intellectium (International Organization affiliated with the United Nations), Mar 2024.
- **Qualified** — International Astronomy and Astrophysics Competition (IAAC), 2021.